

Status. The entire frozen claims pass independent reconstruction and hostile review (AUD077–080), with source and evidence checks complete. The original critical source assertion remains open. This synthesis records the full scope of THM053–055; the complete proofs and evidence are the named round memoranda and audit packets.

Use unit Haar measure on $\mathbb{T}^4$, $c = 4\pi^2$, $\widehat{g}(m) = |m|^{-2}$ for $m \neq 0$, $\widehat{g}(0) = 0$, and $K = -\nabla g$. The original particles start iid Haar, independently of their Brownian drivers, and solve

$$dX_i = \frac{1}{N} \sum_{j \neq i} K(X_i - X_j) dt + \sqrt{2\nu} dW_i, \quad \nu = \beta^{-1}, \quad H_N = \frac{1}{N} \sum_{i < j} g(X_i - X_j).$$

For every $N \geq 2$, the critical window is $0 < \lambda_- \leq \beta/\sqrt{N} \leq \lambda_+ < \infty$. For $m \neq 0$, let $Z_m = N^{-1} \sum_i e^{2\pi i m \cdot X_i}$, $\ell_m = c|m|^2$, $a_m = c + \nu\ell_m$, and $w_m = (1 + |m|)^{10}$. The original response has multiplier $e^{-a_m t}$. Define $J^f(x, y) = K(x - y) \cdot (\nabla f(x) - \nabla f(y))$. For these zero-mean tests, the literal source including both Haar rows is $P_N[J^f] = (2N^2)^{-1} \sum_{i \neq j} J^f(X_i, X_j) + cN^{-1} \sum_i f(X_i)$; its double Haar background is zero. Set

$$S_{N,m}(t) = \sqrt{N} \int_0^t P_N[J^{Q_{i-s}^{\nu} e_m}](X_s) ds, \quad D_{N,m}(t) = \sqrt{N} \{Z_m(t) - e^{-a_m t} Z_m(0)\}.$$

THM054(A) states, for fixed finite T and the entire window,

$$\sup_{t \leq T} \mathbb{E}[|S_{N,m}(t)|^4 + |D_{N,m}(t)|^4 + |\sqrt{N} Z_m(t)|^4] \leq C w_m^4. \quad (1)$$

This is a supremum of deterministic-time expectations.

The new moment mechanism is an exact bounded-kernel identity. For an even real zero-mean bounded kernel h on any compact abelian Haar probability group, put $H^h = N^{-1} \sum_{i < j} h(Y_i - Y_j)$, $\mu_j = \int h^j$, $\tau = \int h^2(h * h)$ and $\chi = \int (h * h)^2$. Then

$$N^4 \mathbb{E}(H^h)^4 = \binom{N}{2} \mu_4 + 18 \binom{N}{3} \mu_2^2 + 36 \binom{N}{3} \tau + 18 \binom{N}{4} \mu_2^2 + 72 \binom{N}{4} \chi. \quad (2)$$

Degree-one labels vanish by centering. The remaining four-edge graphs are one quadrupled edge, two adjacent doubled edges, a doubled-edge triangle, two disjoint doubled edges, and a four-cycle. Their ordering counts give (2).

Apply (2) to $h_L = \min(g, L) - \int \min(g, L)$ at $L = L_0 N^2$. The initial bad event has probability $O(N^{-2})$, while $\mathbb{E}|H^{h_L}|^4 \leq C(N^2 + \log^2 N)$. The true energy identity is $H_t + \mathcal{A}_t = H_0 + \nu c(N-1)t + E_t$, with $\langle E \rangle_t = 2\nu \mathcal{A}_t$ and $H_t \geq -C\sqrt{N}$. Stopping first, the time-zero good-event indicator and the fourth-moment martingale inequality give $\mathbb{E}[\mathbf{1}_{\text{good}} \sup_{t \leq T} |E_t|^4] \leq C$ by absorption. The pointwise source majorant controls the good event. On the bad event, use the exact bounded endpoint and its bounded-bracket observable martingale. This proves (1) without taking an unrestricted moment of the singular energy.

Localization of the exact three-label criterion. Fix a nonzero mode k and time T . Write $a = c + \nu\ell_k$, $\tilde{a} = a - c/N$, $u_t = e^{-\tilde{a}(T-t)}$, $A = e^{-aT}$, and $p_N = (N-1)(N-2)/(2N)$. Let

$$j_k(x, y) = K(x - y) \cdot (\nabla e_k(x) - \nabla e_k(y)) + c(e_k(x) + e_k(y)).$$

The current and initial third-tag correlations are $A_3(t) = \mathbb{E}[j_k(X_1(t), X_2(t))\overline{e_k(X_3(t))}]$ and $B_3(t) = \mathbb{E}[j_k(X_1(t), X_2(t))e_k(X_3(0))]$. At $N = 2$ the three-label quantity is zero. Accepted R25–26 reduce the original source criterion to the real part of

$$R_N^{(3)} = p_N \int_0^T \{u_t^2 A_3(t) - A u_t B_3(t)\} dt. \quad (3)$$

Both weights and the distinction between initial and current tags remain.

Smooth only the observable: $K_r = -\nabla(p_r * g)$ and $d_r = ce^{-r\ell_k}$. The corresponding zero-row kernel $j_{k,r}$ has coefficients

$$\begin{aligned} b_r(m) &= c\{(k \cdot m)a_r(m) - (k \cdot (m+k))a_r(m+k)\} + d_r(\mathbf{1}_{m=0} + \mathbf{1}_{m=-k}), \\ a_r(m) &= |m|^{-2}e^{-cr|m|^2} \quad (m \neq 0), \quad a_r(0) = 0. \end{aligned}$$

Here $b_r(0) = b_r(-k) = 0$, $\sum_m b_r(m) = 2d_r$, and exact empirical deletion gives

$$U_r = \frac{1}{2} \sum_m b_r(m) Z_{k+m} Z_{-m} - \frac{d_r}{N} Z_k.$$

The all-frequency bound $|b_r(m)| \leq C_k |m|^{-2} e^{-cr|m|^2/8}$ implies $\sum_m |b_r(m)| w_{k+m} w_m \leq C_k r^{-11}$. Equation (1), Hölder with exponents 4, 4, 2, and exact overlap counting then yield

$$|R_{N,r}^{(3)}| \leq C \left[N^{-1/2} r^{-11} + N^{-1/4} \sqrt{1 + \log(1/r) + (Nr^2)^{-1}} \right]. \quad (4)$$

The second term uses the actual pair truncated-square estimate and tagged displacement; it uses no positive-time independence.

Set $r_N = r_* N^{-1/24}$, with fixed $0 < r_* \leq r_0$. Replace j_k in (3) by the raw force tail

$$J_k^{>r}(x, y) = (K - K_r)(x - y) \cdot (\nabla e_k(x) - \nabla e_k(y)),$$

and denote the resulting expression by $\mathcal{H}_{N,r}^{(3)}$. The omitted row is explicitly $2p_N(c - d_r) \int_0^T (u_t^2 C_t - A u_t O_t) dt$, where

$$C_t = \frac{N\mathbb{E}|Z_k(t)|^2 - 1}{N-1}, \quad O_t = \frac{N\mathbb{E}[Z_k(t)\overline{Z_k(0)}] - S_t}{N-1}, \quad S_t = \mathbb{E}[e_k(X_1(t))\overline{e_k(X_1(0))}].$$

Equation (1) bounds this row by Cr . Thus (4) proves

$$|R_N^{(3)} - \mathcal{H}_{N,r_N}^{(3)}| \leq CN^{-1/24}. \quad (5)$$

The remaining assertion is precisely $\Re \mathcal{H}_{N,r_N}^{(3)} \rightarrow 0$ for every admitted fixed k, T and critical sequence. An absolute-tail estimate costs $Nr + r^{-1}$ and does not prove it.

Actual first-collision flux for the auxiliary attractive law. For fixed $N \geq 2$ and finite $\nu > 0$, let Y solve the opposite-drift equation with iid Haar start, killed at its maximal collision-free lifetime ζ . Accepted THM052 gives

$$\mathbb{E}[f(Y_t)\mathbf{1}_{t < \zeta}] = e^{-\kappa t} \int f dx, \quad \kappa = c(N-1).$$

The complete configuration gradient is Haar integrable. Tonelli therefore gives the exact first-power action

$$\mathbb{E} \int_0^\zeta |\nabla H_N(Y_s)| ds = \kappa^{-1} \int |\nabla H_N| dx < \infty. \quad (6)$$

Euclidean lifts have an absolutely convergent drift integral and a continuous Brownian limit at the almost surely finite lifetime. Hence $Z = Y_{\zeta-}$ exists. Local continuation would contradict maximality if Z were collision-free.

Let m_{ij} be the probability-Haar law with coordinates i, j fused at a common Haar point and all remaining coordinates independent Haar. Distributionally,

$$\Delta H_N = \kappa dx - \frac{2c}{N} \sum_{i < j} m_{ij}. \quad (7)$$

The coefficient includes both differentiated coordinates, and m_{ij} is a probability measure, not an induced surface measure with a hidden factor. For a globally smooth configuration test, genuine stopped Itô calculus passes to the terminal limit using (6) and bounded observable gradients. Subtracting the surviving Haar part and using (7) identifies the exit measure. Determination of finite Borel measures then gives, for every bounded Borel test ϕ ,

$$\mathbb{E}[\mathbf{1}_{I=\{i,j\}} \phi(\zeta, Z)] = \frac{2c}{N} \int_0^\infty e^{-\kappa t} \int \phi(t, z) dm_{ij}(z) dt. \quad (8)$$

Here almost surely exactly one unordered pair I collides. Indeed every m_{ij} gives all further coincidences measure zero. Equation (8) also holds for nonnegative extended tests and includes $N = 2$. The lifetime is independent of (I, Z) , the pair is uniform, and $Z \mid I = \{i, j\}$ has law m_{ij} . No initial/terminal independence follows.

For $F = N^{-1} \sum_i e_k(x_i)$, define $\Gamma_N = \binom{N}{2}^{-1} \sum m_{ij}$. The exact initial and terminal moments are

$$\int F dx = \Gamma_N[F] = 0, \quad \int |F|^2 dx = \frac{1}{N}, \quad \Gamma_N[|F|^2] = \frac{N+2}{N^2}.$$

The fused pair contributes squared coefficient four, and the other $N-2$ labels contribute one each.

The complete first-collision proof is the Round028 first-collision-flux memorandum. The separate quantitative construction independently recovers the unmarked law before using it. These two ordinary constructions do not replace either fresh independent audit axis. No force-square estimate, postcollision continuation, point-start exponential law or critical cancellation is asserted by (6)–(8).

Quantitative collision-history criterion. Use a, A from the three-label criterion, and put

$$b = c(1 - 1/N) - \nu\ell_k, \quad U = \frac{1}{2N^2} \sum_{i \neq j} j_k(x_i, x_j).$$

The exact attractive generator is $L^+F = bF - U$. Canonically double-centering a radial complex clipping of j_k gives the iid estimate

$$\mathbb{E}_{\text{iid}}|U| \leq u_{N,k} := \sqrt{\frac{(N-1)(1+2C_k \log N)}{2N^3}} + \frac{2C_k(N-1)}{N^2},$$

where the explicit kernel constant C_k is fixed in THM055(B). Only an auxiliary iid estimate is clipped. The killed occupation identity gives the genuine stopped decomposition

$$F(Z) - F(Y_0) = \int_0^\zeta (bF - U)(Y_s) ds + M, \quad \mathbb{E}|M|^2 = \frac{2\nu\ell_k}{N\kappa}.$$

The bound $|F(Z) - F(Y_0)| \leq 2$ and a norm inequality retaining the drift/noise cross term yield

$$N\mathbb{E}|F(Z) - F(Y_0)|^2 \leq \epsilon_{N,k,\nu} := \frac{4N}{\kappa} \left(\frac{|b|}{\sqrt{N}} + u_{N,k} \right) + \frac{2\nu\ell_k}{\kappa}. \quad (9)$$

Conditional on survival to any deterministic T , the entire future segment starts from product Haar. Thus the same bound holds for $N\mathbb{E}[|F(Z) - F(Y_T)|^2 \mid \zeta > T]$. It is $O(N^{-1/2} + \sqrt{1 + \log N}/N)$ uniformly for $0 < \nu \leq \nu_*$. This restarts a future path; it does not condition a past martingale isometry on future survival.

Define the measurable lifetime profile, up to null sets,

$$R(t) = N\mathbb{E}[|F(Y_0) - e^{-at}F(Z)|^2 \mid \zeta = t], \quad B_N(k, T) = \int_0^\infty \kappa e^{-\kappa u} R(T+u) du.$$

Equivalently $B_N = e^{\kappa T} \int_T^\infty \kappa e^{-\kappa t} R(t) dt$. The exponential normalizer remains exact. Let $D_N = N\mathbb{E}_{\text{rep}}|F(X_T) - AF(X_0)|^2$. Full path reversal, (9), and independence of terminal configuration from lifetime give

$$|\sqrt{D_N} - \sqrt{B_N}| \leq A \left[\sqrt{\epsilon_{N,k,\nu}} + \sqrt{(1+2/N) \frac{2a^2}{(\kappa+a)(\kappa+2a)}} \right]. \quad (10)$$

The response-change term inside the square root is exact. The total right side is $O(N^{-1/4})$, uniformly in finite $T \geq 0$ and bounded positive diffusivity. Thus $D_N \rightarrow 0$ if and only if $B_N \rightarrow 0$ along every admitted critical sequence, with the same positive-limsup alternative.

Initial/terminal correlations are the remaining content. The intermediate square equals $v_N + A^2(1 + 2/N) - 2A\Re q_N$, where $v_N = N\mathbb{E}[|F(Y_0)|^2 \mid \zeta > T]$ and $q_N = N\mathbb{E}[F(Y_0)\overline{F(Z)} \mid \zeta > T]$. Conditioning on pair $\{1, 2\}$ leaves these symmetric quantities unchanged; the five label classes have counts $4, 2(N-2), 2(N-2), N-2, (N-2)(N-3)$ before division by N . Their two-time products do not factor. Neither (5) nor (10) decides the original source limit. The old condition $\beta_N \rightarrow 0$, full microscopic subcriticality $\beta_N/\sqrt{N} \rightarrow 0$, and positive critical limit remain separate.